ASK AN ENGINEER

REFERENCE ARCHITECTURES

HOW THIS RUNS, FROM ONE PLANT TO THE WHOLE COMPANY

Every layout below is built from the same two products: SparkBridge Edge at each site, SparkBridge Host in the middle. Add only the modules you need. Every server keeps the systems and databases you already run. Nothing gets torn out to get here.

EIGHT LAYOUTS   SAME TWO PRODUCTS, EVERY TIME   NOTHING RIPPED OUT ON DAY ONE

HOW TO READ A DRAWING   ● SIGNAL ORDER

Green-edged enclosure is an Ignition server. The chips inside are the modules installed on it: green chips are SparkBridge, grey chips are Ignition's own.
Double rule = an MQTT broker. **Dashed outline** = something you already own.
Rows are tiers: field, site servers, brokers, central servers, consumers. The text on the bus between rows is what travels on it.

SHIPPED AND MEASURED   SHIPPED, AWAITING FIELD PROOF   ••• ON THE BENCH

Numbers are from the quiet-box gate of 2026-08-15 and the standby takeover run on a 25-node estate. Where a shape has a limit it is printed under the drawing.

CHOOSE YOUR STARTING POINT

CHOOSE YOUR STARTING POINT

Eight drawings follow, and most readers need one. Find your situation, open that drawing, and skip the rest.

YOUR SITUATION	ARCHITECTURE
One machine or one site, watched from somewhere else	01 the wire
One plant with central operations	02 the plant
10 to 50 distributed sites	03 the data center
Multiple enterprise consumers off one namespace (historian, OPC-UA, cloud)	04 the whole line
History and digital twin	05 the twin
100 to 200+ sites	06 the ultra large estate
Fleet operations: config, identity, audit for one team	07 running it
One Ignition server today, no second site yet	08 one server

The table at the end of the page, Pick by What You Run Today, maps the same drawings from what is already installed.

01 · THE WIRE

ONE MACHINE, WATCHED FROM ANYWHERE

This is the smallest deployment, and every larger one on this page is built from it. A machine on your floor becomes a number your office can see. A change your office makes reaches the machine.

FIELD

PLC

Modbus, Logix, OPC-UA, or one of our drivers

DRIVER

IGNITION SERVER · PLANT

IGNITION

SparkBridge installed here

OPC-UA | LOCAL HISTORY | EDGE

The plant's own full server and database stay. Edge publishes the tag tree as NBIRTH and NDATA and buffers to disk when the link drops

NDATA ↑ · NCMD ↓

BROKER

PROVIDER

SparkBridge Provider. Embedded for a lab, a service beside the server for production.

SPBV1.0/#

IGNITION SERVER · CENTRAL

IGNITION

SparkBridge installed here

HOST | PERSPECTIVE

Rebuilds the tree as a managed provider; a tag write here becomes NCMD to the site

WHAT IT COSTS

SparkBridge Edge at the site, SparkBridge Host at the office. Per gateway. Bought once and kept. Every paid module runs on Ignition's two-hour trial until a key file names it; Reset Trial restarts the clock, and a key lifts the pause within thirty seconds. Support 20%/yr, optional.

WHAT YOU GET	A change on the floor shows up in your office in under a millisecond. Nothing is lost if the connection between them drops.

FOR YOUR ENGINEERS

SERVERS	Two Ignition servers: the plant's existing server (a full Ignition server with its own database, or Ignition Edge at a small onsite plant) and a central server
SPARKBRIDGE	Edge module on the plant server, Host module on the central server. Nothing else, and nothing removed.
ROUND TRIP	p50 0.44 ms, p99 0.97 ms end to end. A 4 h 11 min outage replayed with zero values lost, every one on the timestamp the PLC gave it.
LICENSE	SparkBridge Edge $995 per site gateway, standalone. SparkBridge Host $1,495 per Host gateway, standalone. Bought once.

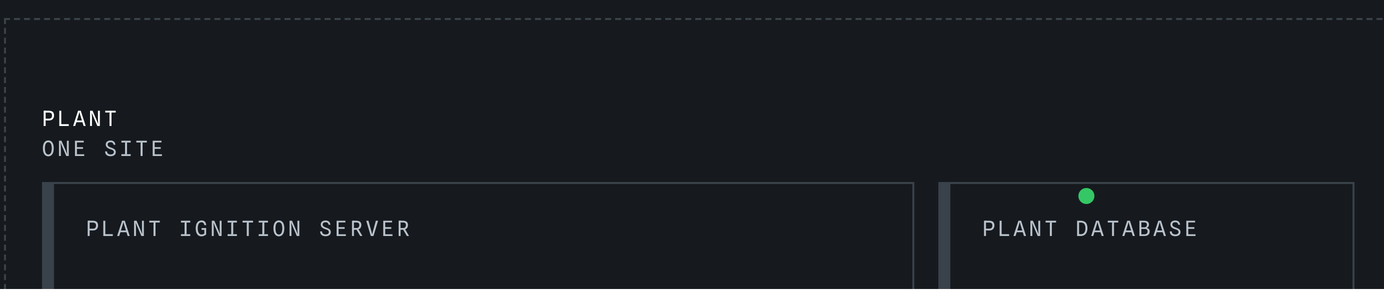
THE LIFE OF ONE CONNECTION   DOWNLOAD

02 · THE PLANT

KEEP THE SERVER YOU ALREADY PAID FOR

Your plant already runs its own server and its own database, with local screens and local alarms. SparkBridge does not replace any of that. It replaces the slow, fragile link that copies data to your office, and hands the plant the whole company's view back.

TODAY · HISTORY WRITTEN TWICE, TAGS FETCHED OVER THE GATEWAY NETWORK



IGNITION

DRIVERS · PERSPECTIVE · ALARMS · HISTORY SPLITTER

Full server. Splitter writes every value to the local database and to the central database.

SQL SERVER · POSTGRES

Local history, alarm journal, local reports

SQL ROWS OVER THE WAN · GAN TAG PROVIDER

HOST

CENTRAL DATABASE

SQL · SQLTH_ PARTITIONS

One connection and one credential per plant. Every plant's rows land here on the central clock.

CENTRAL IGNITION SERVER

IGNITION

GAN REMOTE PROVIDERS · PERSPECTIVE

Reads live tags over the Gateway Network, history from the central database

WITH SPARKBRIDGE · ONE WIRE UP, THE PLANT KEEPS ITS DATABASE

PLANT
ONE SITE

PLANT IGNITION SERVER

IGNITION

SparkBridge installed here

DRIVERS · PERSPECTIVE · ALARMS · LOCAL HISTORY · EDGE · HOST · OPTIONAL

Same server, same screens. Edge publishes the tree; the local history provider keeps writing the plant database. Host, if installed, brings the other plants in as a local provider.

PLANT DATABASE

UNCHANGED

Local history, alarm journal, local reports. Still here when the WAN is not.

NBIRTH · NDATA ↑ · NCMD ↓

BROKER

BROKER

PROVIDER PAIR

SparkBridge Provider, redundant pair. One plant credential, scoped to its own group by SparkID. No database credential leaves the plant.

SPBV1.0/# · SQL ↓ ONCE

HOST

CENTRAL IGNITION SERVER

IGNITION

SparkBridge installed here

HOST | SPARKVAULT | PERSPECTIVE

Live tree from Host; SparkVault writes central history once, on the plant's clock, and serves Perspective trends as a tag-history provider

CENTRAL DATABASE

TIMESCALEDB

One writer. Every plant in one open hypertable, original timestamps, `is_historical` marked on replayed rows.

WHAT STAYS	Your plant server, its screens, its alarms, its database and its local reports. Nothing gets ripped out on day one.
WHAT CHANGES	One reliable wire replaces the slow WAN link to your office, so live data and history arrive from the same source instead of two.
WHAT IT COSTS	Per gateway. Bought once and kept. Every paid module runs on Ignition's two-hour trial until a key file names it; Reset Trial restarts the clock, and a key lifts the pause within thirty seconds. Support 20%/yr, optional.

FOR YOUR ENGINEERS

WHAT STAYS	The plant Ignition server, its drivers, Perspective, alarms, the plant database and the local history provider. The splitter can stay too, writing only its local leg, or be swapped for a plain local provider once the remote leg is off.
WHAT CHANGES	The remote leg of the history splitter and the Gateway Network remote provider are both replaced by one Sparkplug wire. Live tags and central history now come from the same stream.

WHY THE WIRE WINS	The splitter ships SQL rows: one database connection and credential per plant, rows stamped on arrival, and a central schema of sqlth_ partitions nobody wants to query directly. Edge ships values with the timestamp the PLC gave them, buffers to disk through an outage, and SparkVault writes them once into one open table. A 4 h 11 min outage replayed with zero values lost, every one on its original timestamp.
THE PLANT GETS MORE BACK	Install Host on the plant server too and the rest of the estate shows up there as a managed provider: sibling plants, SparkCalc results, corporate setpoints. A plant screen can show the neighbour's line with no Gateway Network link.
CUTOVER ORDER	Install Edge, leave the splitter running both legs, let SparkVault fill in parallel for one history partition, compare, then switch the remote leg off. Nothing is ripped out on day one; the plant's own history is never at risk because it never moves.
SMALL ONSITE PLANTS	A skid, a lift station, a packaging cell with no database of its own runs Ignition Edge instead. Same Edge module, same wire; its history lives at central only, with the Edge server's own disk buffer covering an outage.
LICENSE	Edge $995 per site gateway, standalone, on the plant server. Host $1,495 per Host gateway, standalone, if you also want the estate view there, SparkCalc included. On the Host server: Host $1,495, plus SparkVault $3,995 and Passage $1,995.

Honest limits: reports written against the central sqlth_ tables have to be re-pointed at SparkVault's table or the Timescale hypertable. The plant's own reports do not change. The alarm journal stays in the plant database; alarm state travels as tags, the journal rows do not. Host on a plant server is a full Host license.

SPARKVAULT   STORE AND FORWARD, MEASURED

03 · THE DATA CENTER

ONE VIEW OF EVERY SITE, NOT A DIFFERENT SCREEN FOR EACH

Large operators run several separate servers because one cannot hold every site, and every team has to know which server owns which site. That slows down every decision and every new hire. SparkBridge turns those separate servers into one view, with one Host gateway that grows as you add sites.

TODAY · PER-BACKEND FAN-IN OVER THE GATEWAY NETWORK

PLANTS
40 IGNITION SERVERS, 40 DATABASES

PLANTS 01 TO 14
IGNITION + LOCAL DB
GAN REMOTE | SPLITTER

PLANTS 15 TO 28
IGNITION + LOCAL DB
GAN REMOTE | SPLITTER

PLANTS 29 TO 40
IGNITION + LOCAL DB
GAN REMOTE | SPLITTER

GAN PER PLANT · SQL ROWS PER PLANT

BACKENDS
ONE PER SLICE

BACKEND 1
IGNITION
Holds sites 01 to 14. Full at this size.

BACKEND 2
IGNITION
Holds sites 15 to 28

BACKEND 3
IGNITION
Holds sites 29 to 40

CENTRAL DATABASE
40 SPLITTER CONNECTIONS
Every plant holds a credential to it

REMOTE TAG PROVIDERS · THREE OF THEM

APPS

APPLICATION SERVERS

IGNITION · PERSPECTIVE

Every screen has to know which backend owns which site. Add a site, touch every consumer.

WITH SPARKBRIDGE · ONE NAMESPACE, A POOL OF HOSTS

PLANTS
40 IGNITION SERVERS, 40 DATABASES

PLANTS 01 TO 40

IGNITION + LOCAL DB

SparkBridge installed here

DRIVERS | LOCAL HISTORY | EDGE

Each plant keeps its database and publishes its own group; the splitter's remote leg and the GAN link both retire

SMALL ONSITE PLANTS

IGNITION EDGE

SparkBridge installed here

EDGE

Skids and cells with no database; history lives at central

NBIRTH · NDATA · QOS 1

BROKERS
DATA CENTER

BROKER A

PROVIDER

SparkBridge Provider. Retains STATE and birth certificates

BROKER B

PROVIDER

SparkBridge Provider, paired with A; edges fail over between them

$SHARE/HOSTS/… · SPBV1.0/#

HOST
IGNITION HOST POOL

CENTRAL 1 · PRIMARY

IGNITION

SparkBridge installed here

CENTRAL 2 · STANDBY

IGNITION

SparkBridge installed here

HOST · 4 CONSUMERS

SPARKVAULT

Owns the host id; publishes STATE, sends commands. SparkVault writes central history once.

HOST · STANDBY

Same state, same id, commands fenced until it claims

ONE MANAGED PROVIDER · EVERY SITE

APPS

APPLICATION SERVERS

IGNITION · PERSPECTIVE

One remote provider, one tree, every plant. Add a plant and it appears.

CENTRAL DATABASE

TIMESCALEDB

One writer, no plant credentials, original timestamps

WHAT YOU GET	Forty plants, one screen. Add a plant and it shows up automatically, no rework for the team watching it.
UPTIME	If the primary Host gateway fails, a standby takes over in about a quarter of a second. Nobody at any site notices.
WHAT IT COSTS	Per gateway. Bought once and kept. Every paid module runs on Ignition's two-hour trial until a key file names it; Reset Trial restarts the clock, and a key lifts the pause within thirty seconds. Support 20%/yr, optional.

FOR YOUR ENGINEERS

SERVERS	40 plant Ignition servers with their 40 databases, untouched; 2 brokers; 2 central Ignition servers in a host pool; one central database; your existing application servers
SPARKBRIDGE	Edge on every plant server; Host and SparkVault on every server in the central pool

CAPACITY	At least 340,000 updates/s sustained through to visible tags on one central server; 480,000 offered without bending. Four shared consumers ingest about 20 times what one does. 500,000 metrics resident in under 4 GiB.
TAKEOVER	Standby claims the host id in 244 ms end to end, 5.7 ms mechanical, with zero rebirths. The sites never notice.
SIDEWAYS	Add a central server to the pool for more consumers, or partition the namespace by group across servers with ordering preserved. Both are live changes, no rebirth wave.
LICENSE	Edge $995 per site gateway, standalone. Host $1,495 per pool gateway, standalone.

Honest limits: standby caps you at two writers for at most the claim-stale window, not zero. Shared consumers trade per-edge ordering for throughput; partition by group if ordering matters. Run the standby against one broker server, since convergence is per broker.

PERFORMANCE QUANTIFIED   STANDBY AND SCALE-OUT

04 · THE WHOLE LINE

EVERY TOOL YOUR COMPANY NEEDS, OFF ONE NAMESPACE

This is the data center from section 03 with every optional module turned on: your history, your security audit, your reporting tools, your cloud warehouse, all reading the same feed instead of five.

FIELD

MODERN
PLCS

LEGACY
RTUS · 505 RACKS

REMOTE

Logix, Modbus, OPC-UA

IEC-104, DNP3, TI 505

WELLHEADS · LIFT STATIONS

Cellular and satellite

DRIVERS

SITES
PLANT AND EDGE SERVERS

PLANTS
IGNITION + LOCAL DB
SparkBridge installed here

OPC-UA · LOCAL HISTORY · EDGE

Full servers, as in 03

SMALL ONSITE PLANTS
IGNITION EDGE
SparkBridge installed here

EDGE

UTILITY SITES
IGNITION EDGE
SparkBridge installed here

IEC-104 · DNP3 · TI 505 · EDGE

METERED SITES
IGNITION EDGE
SparkBridge installed here

MODBUS · EDGE · SPARKLINK

NBIRTH · NDATA ↑ · NCMD ↓ · SIGNED CONFIG ↓

BROKERS
DATA CENTER

BROKER A
PROVIDER
SparkBridge Provider. Rules from SparkID: only the host id may command, each site scoped to its own group

BROKER B
PROVIDER
SparkBridge Provider, paired with A

NOT A GATEWAY
SPARKID
Generates credentials.xml, extension config, NATS ACLs, standby bindings; rotates with overlap

SPBV1.0/# · $SHARE · STATE · CLAIM

HOST
IGNITION HOST POOL

CENTRAL 1 · PRIMARY
IGNITION

SparkBridge installed here

HOST · 4 CONSUMERS | SPARKVAULT | SPARKCALC | PASSAGE | SPARKINJECT | SPARKFLOW | FLEETOPS CONSOLE | PERSPECTIVE

One managed provider with every site. Vault stores it, Calc derives from it under its own group, Passage serves it as OPC-UA, Inject and Flow send it out.

CENTRAL 2 · STANDBY

IGNITION

SparkBridge installed here

HOST · STANDBY | SPARKVAULT

Same state; takes the id in 244 ms

SQL · OPC-UA · NATS · STREAMING

CONSUMERS

VIA SPARKVAULT

TIMESCALEDB

Open history on the plant's clock; replay any day

VIA PASSAGE

SCADA · MES · DCS

Browse the namespace over OPC-UA

VIA SPARKFLOW

NATS · THE COMPANY

Subjects, last value, model bucket

VIA SPARKINJECT

SNOWFLAKE

Streaming, exactly once

CLAMPED ACROSS ALL OF IT **SENTINEL**

Audits the brokers and the host from outside with read-only probes, reports what is exposed, and feeds the Compliance binder. Never in the data path.

WHAT YOU GET

One license for every site, one office server reporting on security, cost, history and cloud delivery. No custom integration project to reach any of it.

WHAT IT COSTS

Per gateway. Bought once and kept. Every paid module runs on Ignition's two-hour trial until a key file names it; Reset Trial restarts the clock, and a key lifts the pause within thirty seconds. Support 20%/yr, optional.

FOR YOUR ENGINEERS

SERVERS	Plant Ignition servers with their local databases, Ignition Edge at small onsite plants, 2 brokers, 2 central Ignition servers, a SparkID workstation, your database, NATS and warehouse
ON EACH PLANT OR EDGE SERVER	SparkBridge Edge, plus our IEC-104 / DNP3 / TI 505 drivers where the rack needs them, plus SparkLink where the link is metered
ON EACH CENTRAL SERVER	SparkBridge Host, SparkVault, SparkCalc, Passage, SparkInject, SparkFlow, FleetOps. All are Ignition modules on the same server; none needs a box of its own.
OFF THE SERVERS	SparkID and Sentinel are tools you run from a workstation against the estate; nothing installs on a gateway for them
LICENSE	Per site gateway: Edge $995 standalone, SparkLink included. Per Host gateway: Host $1,495 standalone, SparkCalc and SparkID included. Other modules on Host: Passage $1,995, Sentinel and FleetOps $4,995 each, SparkInject $2,995 per connector family, SparkVault $3,995. Provider $1,995 per gateway, managed broker and policy layer, works with Edge and Host or with any conformant Sparkplug host. SparkFlow $995, SparkSNMP $695 and drivers $695 each carry their own license. Per gateway, bought once and kept. Every paid module runs on Ignition's two-hour trial until a key file names it; Reset Trial restarts the clock, and a key lifts the pause within thirty seconds. Support 20%/yr optional.

THE PRODUCT LINE   EDGE AND HOST PRICING

05 · HISTORY AND THE TWIN

EVERY DAY, ON RECORD AND REPLAYABLE

Both your history and your live model live on the central server. Neither touches a site. History is kept on your own clock, so an outage never leaves a hole and any past day can be replayed. The live model is the same data, always current, for the tools that need it.

BROKERS

LIVE ESTATE

PROVIDER PAIR

SparkBridge Provider. Every site's NBIRTH, NDATA and templates

REPLAY NAMESPACE

REPLAY/<GROUP>

SparkVault re-publishes history here as its own edge, never under a site's id

SPBV1.0/# IN · REPLAY/ OUT

HOST
IGNITION HOST POOL

CENTRAL 1 · PRIMARY

IGNITION

SparkBridge installed here

HOST | SPARKVAULT | SPARKFLOW | PASSAGE | SPARKCALC | TAG HISTORY | PERSPECTIVE

SparkVault is also an Ignition tag-history provider, so Perspective trends read estate history with no new tooling. SparkCalc backfills derived values over that history as historical, not live.

SQL ↓ · KV PUTS · OPC-UA

THE TWIN
THREE FACES

HISTORY

TIMESCALEDB

Hypertable, original timestamps, `is_historical` preserved through store and forward. Replay any day at 1x or faster.

LIVE MODEL

NATS · UNS-LAST · UNS-MODEL

Last value per metric, plus the structure: templates and birth properties per node and device, read at `GET /model`

STRUCTURED FACE

OPC-UA ADDRESS SPACE

group / edge / device / metric, live subscriptions, allow-listed writes down

SIMULATION, MES, ANALYTICS, AI

USES

REPLAY

WALK AN INCIDENT

Re-run the hour it happened through the real screens and alarms

SIMULATE

TEST BEFORE CUTOVER

Play history at speed against a candidate host or a new dashboard

MODEL

COMPANY SYSTEMS

Subscribe to the structure, not just the numbers

WHAT YOU GET	Every site's history, kept on your own clock, any past day replayable. A live model of the plant for the tools that want one, always current.
WHAT IT COSTS	Per gateway. Bought once and kept. Every paid module runs on Ignition's two-hour trial until a key file names it; Reset Trial restarts the clock, and a key lifts the pause within thirty seconds. Support 20%/yr, optional.

FOR YOUR ENGINEERS

SERVERS	The central pool from 02 and 03. No new Ignition server; the database and NATS are yours.
HISTORIAN	SparkVault on the central server; each plant's own database keeps its local history exactly as before. History for every site in one open table, on the plant's clock. 40,035 updates/s replay on loopback; a 4 h 11 min outage backfilled with zero loss.
TWIN	SparkFlow's last-value and model buckets plus Passage's OPC-UA address space plus SparkVault history. Nothing is copied into a second tool; each face reads the same namespace.
DISCIPLINE	Replayed and computed values always publish under their own group with their own sequence, so a simulation or a derived number can never be mistaken for a live measurement.
LICENSE	SparkVault $3,995 on the central server; Passage $1,995; SparkCalc included with Host. SparkFlow $995 carries its own license. SparkBridge Provider handles the broker layer for this drawing; any conformant Sparkplug broker also works.

SPARKVAULT SPARKFLOW PASSAGE SPARKCALC

06 · THE ULTRA LARGE ESTATE

TWO HUNDRED PLANTS, ONE NAMESPACE, NO CEILING

Past a certain size, no single server can hold everything. SparkFlow, its own license, is how the largest companies scale past that ceiling: each region keeps what its own operators watch, and the whole company still reads one namespace with nothing left out.

PLANTS
200 IGNITION SERVERS

REGION EAST · PLANTS 001 TO 070

IGNITION

SparkBridge installed here

DRIVERS | LOCAL HISTORY | EDGE

Full plant servers with their own databases, as in 03

REGION CENTRAL · PLANTS 071 TO 140

IGNITION

SparkBridge installed here

DRIVERS | LOCAL HISTORY | EDGE

REGION WEST · PLANTS 141 TO 200

IGNITION EDGE · IGNITION

SparkBridge installed here

DRIVERS | EDGE

Mixed: full plants and small Edge sites

NBIRTH · NDATA ↑ · NCMD ↓ · PER REGION

BROKERS
OT TIER, PER REGION

EAST

PROVIDER PAIR

SparkBridge Provider. Sparkplug, MQTT 5,

HOST

PROVIDER PAIR

SparkBridge Provider. Sparkplug, MQTT 5,

WEST

PROVIDER PAIR

SparkBridge Provider. Sparkplug, MQTT 5,

$share. Rules from SparkID.

$share. Rules from SparkID.

$share. Rules from SparkID.

$SHARE/HOSTS · SPBV1.0/#

REGIONAL
IGNITION HOST POOLS

EAST POOL · PRIMARY + STANDBY

IGNITION

SparkBridge installed here

HOST · 4 CONSUMERS

SPARKFLOW · SEPARATE LICENSE

SPARKCALC

PERSPECTIVE

Tags: the ~150,000 points East operators watch. Stream: all 1.8 million points East decodes.

CENTRAL POOL

IGNITION

SparkBridge installed here

HOST

SPARKFLOW · SEPARATE LICENSE

Same shape. Namespace partitioned by group; ordering kept per edge.

WEST POOL

IGNITION

SparkBridge installed here

HOST

SPARKFLOW · SEPARATE LICENSE

UNS.<GROUP>.<EDGE>.<DEVICE>.<METRIC> · EXACTLY ONCE

BACKBONE
NATS, APP TIER

PER REGION

NATS · JETSTREAM

Stream TELEMETRY on uns.>, KV uns-last (value, ts, quality per point), KV uns-model (structure from birth certificates). Each region owns its JetStream domain and keeps acking with the hub down.

COMPANY

NATS HUB

Sources TELEMETRY_EAST, TELEMETRY_CENTRAL, TELEMETRY_WEST from the regions over TLS leaf links. Regions dial out; the hub never dials in. Msg-Id survives the mirror, so a hub-side historian dedups exactly like a regional one.

DURABLE CONSUMERS · KV READS · UNS.CMD ↓

COMPANY

CONSUMERS

CORPORATE HISTORIAN

IGNITION

SparkBridge installed here

SPARKVAULT · JETSTREAM INGEST

SPARKINJECT | PERSPECTIVE

Vault drains the hub stream on a durable consumer into one Timescale history for all 200 plants; Inject streams the same to the warehouse

CORPORATE PERSPECTIVE

IGNITION · DISPATCHER

SparkBridge installed here

SPARKFLOW DISPATCHER | PERSPECTIVE

Screens bind to points on demand from uns-last with no tag behind them. On the bench: the SDK hook is stubbed, the dispatcher core is written.

DIRECT

ANALYTICS · AI · MES

Subscribe to a subject filter, open with the last value of everything, replay any window in order

WHAT YOU GET	Every plant in the company in one view, with no ceiling on how large that view can grow.
MEASURED	7.25 million messages delivered across a real network, in order, exactly once, zero lost through a physical link cut.
WHAT IT COSTS	SparkFlow carries its own license, per gateway, bought once and kept.

FOR YOUR ENGINEERS

SERVERS	200 plant Ignition servers, 3 broker pairs, 3 regional host pools of 2, one corporate historian server, one corporate Perspective server, 4 NATS servers. The plants are the drawing in 03 repeated.
WHERE THE CEILING GOES	No Ignition server anywhere holds five million tags. Each regional pool holds the slice its operators watch as tags and streams everything it decodes. The company namespace lives in NATS: one subject per point, a last value and a model for each, no tag license behind any of it.
MEASURED	7.25 million messages across a real network at 48,000/s, 7.25 million received, in order, exactly once. A 60 s physical link cut at a leaf: 447

	held, 447 reconciled at the hub, zero lost.
COMMANDS	Routed rather than mirrored. `uns.cmd.<group>.<edge>…` request and reply crosses the leaf link natively; only the region's bridge subscribes to its own command subtree, and only the issuers SparkID names may publish there. A region that cannot confirm a write times the requester out rather than reporting success.
TIERS, HONESTLY	SparkBridge Provider stays the OT tier: Sparkplug conformance, MQTT 5, shared subscriptions, TCK, all measured there; any conformant Sparkplug broker also works. NATS sits north of it as the backbone. SparkFlow is the certified translation between the two, not a replacement for either.
LICENSE	SparkFlow $995 per Host gateway running it: three regional pools and the corporate historian, each its own license, on top of anything else on that server. Edge $995 per plant, Host $1,495 per pool server. NATS is open source and yours.

> Honest limits: a single region, one NATS domain, one host pool, SparkFlow, SparkVault ingest, the leaf partition, is shipped and gated. The multi-region hub with per-region accounts is a design built on those gates. Cross-domain sourcing across an account boundary is the one spike we still owe before a customer install, and that's why the hub is drawn dashed. The Perspective dispatcher is on the bench.

SPARKFLOW SPARKVAULT SPARKID

07 · RUNNING IT

FORTY SITES, RUN AND PROVEN FROM ONE CHAIR

This is what running the estate from section 03 looks like day to day. One team pushes a change to every site at once. Access is generated instead of typed by hand. An outside audit checks the whole thing on demand.

OPS
WORKSTATION

NOT A GATEWAY

FLEET PUBLISHER

P-256 signing key; site documents in git; publishes signed, sequenced, retained config

NOT A GATEWAY

SPARKID

One declaration of who may read and who may command; emits broker rules, NATS ACLs, standby bindings; dry-run diff against the live broker

NOT A GATEWAY

SENTINEL

Anonymous connect, command-topic reachability, passive capture. Reports and scores; never remediates.

CONFIG ↓ · RULES ↓ · PROBES

BROKERS

BROKER PAIR

PROVIDER

SparkBridge Provider. sparkbridge/config/<group>/<edge> retained; credentials and topic scoping from SparkID

CONFIG ↓ · NODE INFO IN NBIRTH ↑

SITES AND CENTRAL
IGNITION SERVERS

SITES 01 TO 40

IGNITION EDGE

SparkBridge installed here

EDGE · TRUSTED KEY

Verifies, refuses replay below its sequence floor, applies without restart

CENTRAL POOL

IGNITION

SparkBridge installed here

HOST | FLEETOPS CONSOLE

Module version, config version, buffered bytes and drops for every site on one grid; drift and cert expiry ledger

WHAT YOU GET

A change reaches every site at once, from one place, verified before it applies. Access is generated, not typed by hand, so nobody has to remember who can do what.

WHAT IT COSTS

Per gateway. Bought once and kept. Every paid module runs on Ignition's two-hour trial until a key file names it; Reset Trial restarts the clock, and a key lifts the pause within thirty seconds. Support 20%/yr, optional.

FOR YOUR ENGINEERS

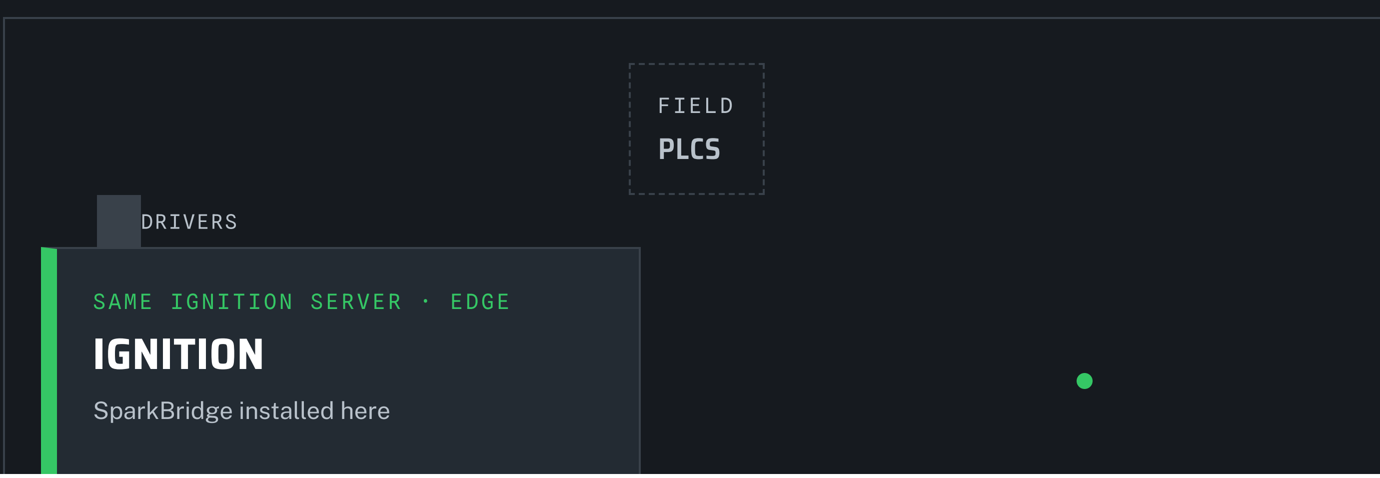
SERVERS	No new Ignition servers. One ops workstation runs the three tools.
TRUST	A site with no trusted key subscribes to nothing. The signing key is a root credential for the estate; it lives on the workstation, never on a gateway.
PROOF	A Sentinel run becomes an auditor-ready binder mapped to NERC CIP and IEC 62443 through the Compliance add-on
LICENSE	FleetOps $4,995, SparkID included with Host, Sentinel $4,995; Sentinel compliance pack $2,495 add-on.

FLEETOPS SPARKID SENTINEL

08 · THE SMALL END

ONE PLANT, ONE SERVER, ROOM TO GROW

Everything above also fits on a single server for a single plant. Start here. The day you add a second site, nothing about the shape changes: it becomes site one of the data center.



DRIVERS LOCAL HISTORY EDGE

Publishes plant data to the required Provider broker. The physical server stays one enclosure.

SPARKPLUG

BROKER LAYER

PROVIDER

Embedded Moquette for a single node, or a HiveMQ CE companion service for MQTT 5 and shared subscriptions.

LOCAL SPARKPLUG

SAME IGNITION SERVER · HOST

IGNITION

SparkBridge installed here

HOST SPARKVAULT PASSAGE
SPARKCALC PERSPECTIVE

Consumes through Provider, keeps history, serves OPC-UA and computes in one enclosure. Point it at a central broker later and it becomes site 01 of the data center.

SQL · OPC-UA

HISTORY

TIMESCALEDB

OTHER SYSTEMS

OPC-UA CLIENTS

WHAT YOU GET

One server does everything a plant needs: moves the data, keeps the history, serves it out. It grows into the data center in section 03 with no rework.

WHAT IT COSTS	Per gateway. Bought once and kept. Every paid module runs on Ignition's two-hour trial until a key file names it; Reset Trial restarts the clock, and a key lifts the pause within thirty seconds. Support 20%/yr, optional.

FOR YOUR ENGINEERS

SERVERS	One Ignition server
BROKER	Embedded through the Provider module for a lab; SparkBridge Provider as a service beside the server for production, which is also what makes the later move to 03 a configuration change. Any conformant Sparkplug broker also works.
LICENSE	Edge $995 per site gateway, standalone, SparkLink included. Add Host $1,495 and SparkVault $3,995 on this server only if it is also acting as the Host gateway with its own database.

PRICING DOWNLOAD

WHICH DRAWING IS MINE

PICK BY WHAT YOU RUN TODAY

YOU RUN TODAY	YOUR DRAWING	WHAT CHANGES
One Ignition server, one plant	08 one server	Install the modules; nothing else moves
A site server and a central server on the Gateway Network	01 the wire	Edge on the site, Host at the center, a broker between; the GAN link retires
Full plant servers with local databases and a history splitter	02 the plant	Plant server and database stay; the splitter's remote leg becomes the

YOU RUN TODAY	YOUR DRAWING	WHAT CHANGES
to a central database		Sparkplug wire and SparkVault writes central history
Several Ignition backends because one cannot hold every site	03 the data center	Backends become a host pool behind one broker pair; consumers read one tree
The above, plus a historian, an OPC-UA face or a cloud warehouse to feed	04 the whole line and 05 the twin	Modules on the central pool; no new servers
Hundreds of plants and millions of points, more than any Ignition server can hold as tags	06 the ultra large estate	Regional host pools hold what operators watch; SparkFlow streams everything onto NATS; the company reads one namespace
Forty sites and one team to run them	07 running it	An ops workstation, signed config, generated identity, outside audit

What is not on this page. No drawing mixes another vendor's Sparkplug modules with ours. That pairing has never run on our bench, so we are not going to imply compatibility we have not measured.

DRAW YOURS WITH US

Start from the written 40-site reference architecture (PDF) or request a review of your own estate.

Send us the server list, the site count, and what has to read the data. You get back one of these drawings with your server names on it. Try it on one site: every paid module runs on Ignition's two-hour trial until a key file names it, and a key lifts the pause within thirty seconds.

CONTACT

SEE THE PRODUCTS